Machine Learning – Exercise 0: Dataset Description

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1 Dataset Selection

We selected the Cylinder Bands and Page Blocks datasets from OpenML because they differ significantly in their characteristics.

The number of observations differs by approximately a factor of ten, with 540 observations in Cylinder Bands and 5,473 in Page Blocks. The number of input attributes also differs considerably, with 39 attributes in Cylinder Bands compared to only 10 in Page Blocks.

Another important difference is the number of target classes. Cylinder Bands is a binary classification dataset with two classes: *band*, indicating the occurrence of cylinder banding during the printing process, and *noband*, indicating its absence.

Page Blocks, on the other hand, has five target classes representing different types of document elements: text, horizontal lines, pictures, vertical lines, and graphics.

The attribute types also differ. Cylinder Bands contains a mixture of numerical and categorical attributes, whereas Page Blocks contains exclusively numerical input attributes.

Furthermore, Cylinder Bands contains missing values across several attributes, while Page Blocks has no missing values.

2 Dataset Characteristics

Characteristic	Cylinder Bands	Page Blocks
Observations	540	5,473
Input attributes	39	10
Target classes	2	5
Input attribute types	Mixed	Numerical
Missing values	Yes	No

Table 1: Comparison of the selected datasets.

3 Target Class Distributions

The target class distributions also show clear differences between the datasets.

Cylinder Bands has a relatively balanced distribution, with 57.8% of observations belonging to the *noband* class and 42.2% to the *band* class.

Page Blocks, however, has a strongly imbalanced class distribution. The text class accounts for approximately 89.8% of all observations, while the other four classes together represent only about 10.2%. The smallest class contains just 28 observations (0.5%).

This difference is important because class imbalance can affect classification performance and the interpretation of evaluation metrics.

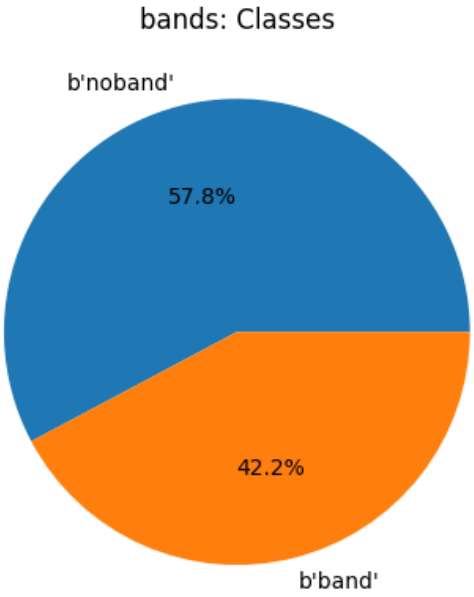
4 Missing Values

Cylinder Bands contains missing values in several attributes, while Page Blocks has no missing values at all.

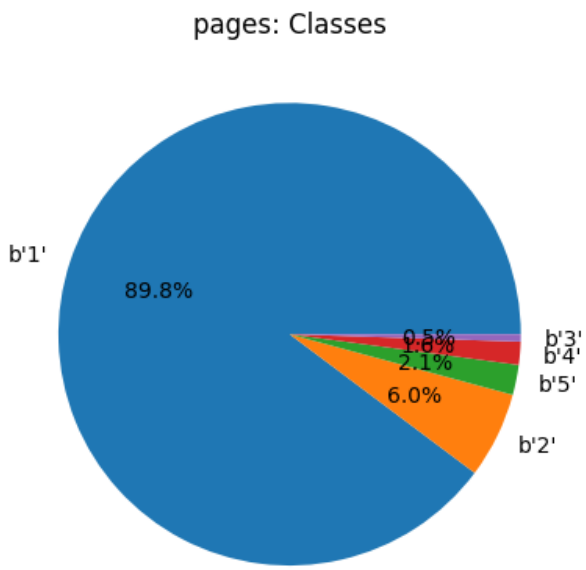
In Cylinder Bands, 28 of the 39 input attributes contain missing values. The attribute `paper_mill_location` has the highest number of missing values, with 156 out of 540 observations (28.9%). Other attributes, such as `blade_pressure`, also have a relatively high number of missing values.

These missing values need to be considered during preprocessing. For numerical attributes, missing values could be replaced by the median, while for categorical attributes, the most frequent category could be used. Alternatively, attributes with too many missing values could be excluded.

Since Page Blocks has no missing values, no missing-value imputation is required for this dataset.



(a) Cylinder Bands



(b) Page Blocks

Figure 1: Target class distributions.

5 Numerical Attributes and Preprocessing

The numerical attributes also differ considerably between the two datasets, both in their value ranges and distributions.

For Cylinder Bands, attributes such as `press_speed` have values ranging from 0 to 2600, while others, such as `roughness`, range only from approximately 0.06 to 1.25. The dataset also contains numerical and categorical attributes, meaning that categorical variables may need to be encoded before applying certain machine learning algorithms.

In Page Blocks, the differences in numerical ranges are even more noticeable. For example, `area` ranges from 7 to 143993, while `p_black` ranges from approximately 0.05 to 1. Some attributes also show strongly skewed distributions and extreme values, as illustrated by the logarithmic boxplot.

Therefore, feature scaling could be useful for both datasets, especially for algorithms that are sensitive to differences in numerical scales. In addition, extreme values should be investigated before deciding whether any further transformations are necessary.

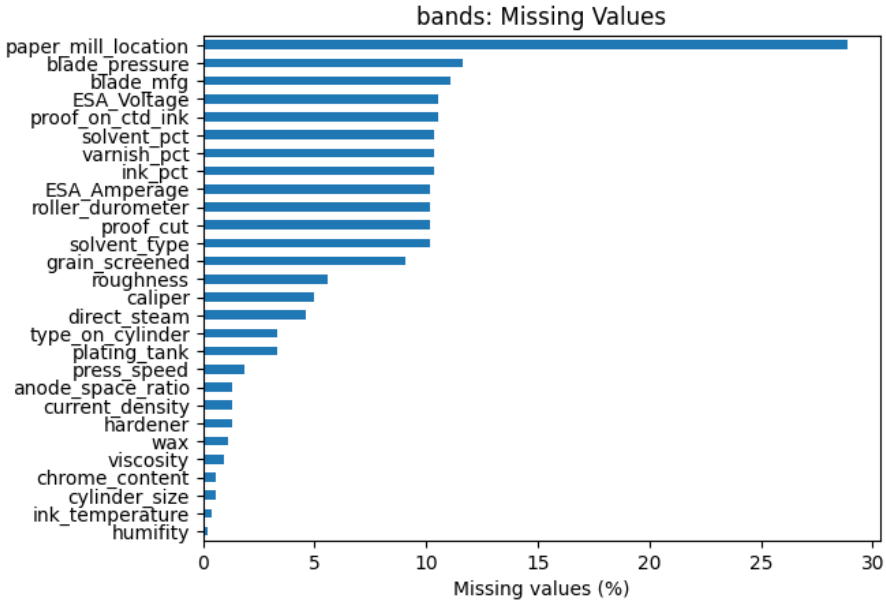
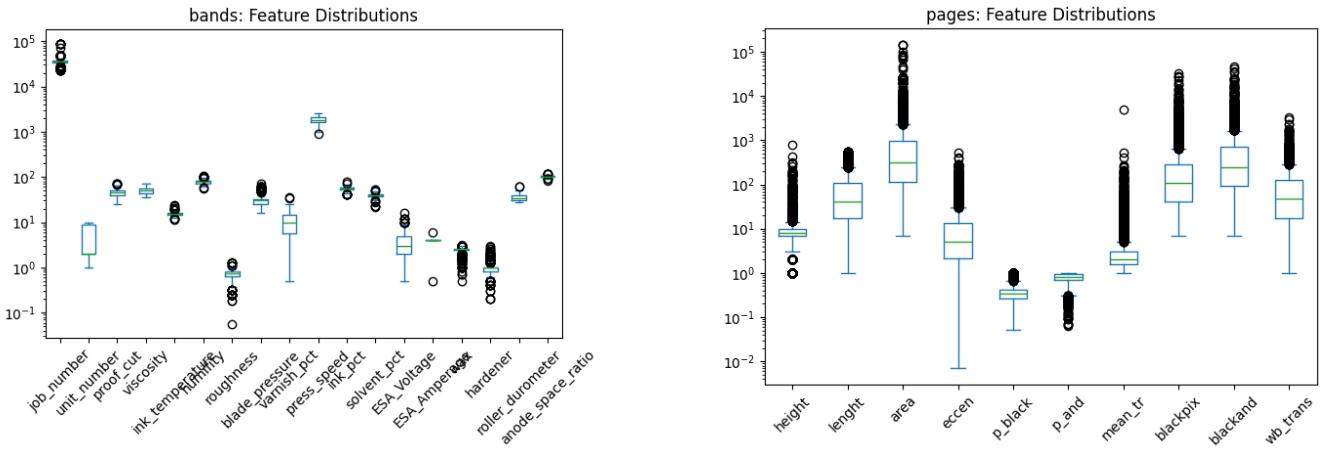


Figure 2: Percentage of missing values in Cylinder Bands.



(a) Cylinder Bands

(b) Page Blocks

Figure 3: Numerical feature distributions on a logarithmic scale.